IMPERIAL COLLEGE LONDON

EE4-50
EE9-FPN2-10

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART IV: MEng and ACGI

CORRECTED COPY

SUSTAINABLE ELECTRICAL SYSTEMS

Thursday, 9 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question 1 & 4 in Booklet A, Questions 2 & 3 in Booklet B.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : B. Chaudhuri, G. Strbac
Second Marker(s) : G. Strbac, B. Chaudhuri

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Answer ALL Questions

1. Question on Wind and Photovoltaic Power

a) The hourly wind speed (in m/s) profile across the year at a particular location is modelled by Weibull probability distribution where the probability of experiencing wind speed less than v_w is given by:

$$\Phi(v_w) = 1 - e^{-\left(\frac{v_w}{8}\right)^{2.2}}$$

At this location, a 8 MW direct-drive wind turbine with a permanent magnet synchronous generator (PMSG) is installed. The wind turbine parameters are given in Table 1.1 and the density of air can be assumed to be 1.22 kg/m^3.

Length of turbine blades	r	85 m
Maximum power coefficient	C_{p-max}	0.5
Optimum tip speed ratio	λ_{opt}	8
Cut-in speed	v_{w-in}	4 m/s
Cut-out speed	v_{w-out}	25 m/s

Table 1.1: Parameters of the wind turbine

i) Calculate the rated wind speed (in m/s)

[1]

ii) What is the probability of encountering wind speeds higher than the rated speed?

[1]

iii) What is the probability of encountering wind speeds in the region where the wind turbine operates in maximum power extraction mode by adjusting the turbine rotational speed?

[1]

iv) Determine the range of rotational speed (in rpm) of the turbine required to extract maximum power from wind.

[2]

v) How many poles are needed for the PMSG if the minimum allowable stator frequency is 10 Hz?

[2]

b) The declination angle of earth on a particular day is 18°. For this particular day and a specific location with a latitude of 50°N:

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i) Calculate the elevation angle (in degrees) and the air mass (AM) at solar noon.

[2]

ii) Determine the optimum tilt angle (in degrees with respect to the horizontal) of a solar panel at solar noon in order to maximise the direct irradiance.

[2]

c) Partial shading of PV modules or PV arrays are known to be problematic.

i) State two problems associated with shading of a few PV cells within a PV module with series connected PV cells.

[2]

ii) Mention the role of string diodes in a solar PV array with parallel strings of PV modules and state the drawback of using the string diodes.

[2]

d) A solar PV module with 96 PV cells has a Watt Peak rating of 350 W corresponding to the standard test condition (STC). The PV cells are connected in series with a bypass diode across groups of 32 of them. The open circuit voltage and short circuit current of each cell under STC is 0.7 V and 6.5 A, respectively. The efficiency of the PV cells is 16.5% under STC.

A PV array is formed with 3 parallel strings. Each string has 4 PV modules in series with a string diode. The total current from the 3 strings is 18 A for maximum power output at STC.

Neglecting other losses within the PV module apart from that in the PV cells, calculate the following assuming STC for all the PV cells:

i) Fill factor of each PV cell.

[1]

ii) Surface area of each PV cell in cm^2.

[1]

iii) Voltage at which the PV cells produce maximum power.

[1]

iv) Power (in kW) generated by the PV array if one PV module within a particular string is completely shaded.

[2]

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2. Question on Distributed Generation, Voltage Control and Security of Supply

Part A

As indicated in the network diagram presented in Figure 2.1, there is a need to assess the capacity of wind farm that could be connected to the network in Node C, under different voltage control strategies in order to maintain voltage variations in the network within ±6%.

Capacity of both 11 kV underground cables is 900 Amps with resistivity of 0.065 Ohms/km and reactivity of 0.095 Ohms/km.

Note that the maximum and minimum power demands at Node B are 20MW and 10MW respectively, and that power factor is 0.96 inductive.

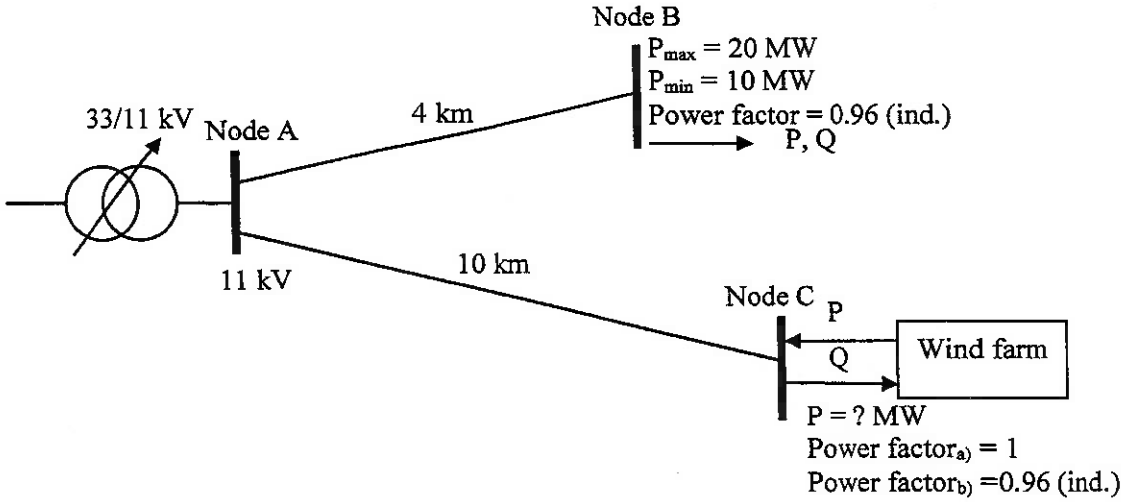


Figure 2.1 – Network topology

a) Determine the maximum capacity of the wind farm that could be connected to Node C if the On Load Tap Changing (OLTC) transformer maintains the voltage at Node A at 1.03 p.u., while the wind farm operates at unity power factor. [4]

b) In case that the wind farm can import reactive power and operate at the power factor of 0.96 inductive, determine the amount of additional capacity of wind generation that can be connected to Node C. [3]

c) Assuming that the voltage at Node A is optimised by OLTC, determine the maximum capacity of the wind farm that can be connected to Node C, while the wind farm operates at the power factor of 0.96 inductive. [3]

Part B

a) Explain the fundamental approach used to quantify the contribution of "non-network technologies" to network security. [3]

b) Consider 30 MW generation plant that consists of three identical generation units, with technical availability of each unit being 86%. Determine the contribution of the plant to security of supply

(i) During peak season when all units are operational; and

(ii) During off-peak season when two units are operational (as one-by-one would be on maintenance).

Normalised load duration curves for peak and off-peak seasons are shown in Figure 2.2. Note that the duration of Peak season is 40% while the duration of Off-peak season is 60% of the year and that each of the seasons is represented with two demand levels (with specified durations of high and low demand periods in each season).

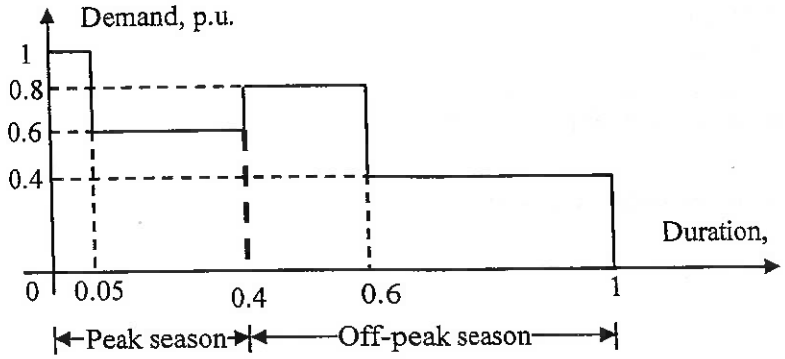


Figure 2.2 Normalised load duration curve

When calculating the contribution to security of supply of the generation plant, scale the demand in each season to the total generation plant capacity available in that season.

[7]

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3. Question on Low Carbon Heat

a) Give a brief description of the factors that determine the demand for space heat in a domestic building.

[3]

b) The figure below shows the total daily GB demand for domestic space and water heating plotted against average daily external temperature. Assuming:

- 30 million domestic buildings
- Average building floor area of $100m^2$
- Space heating period is 200 days
- Water heating period is 365 days
- Average daily external temperature for the year of 7°C
- Average building internal temperature for the year of 17°C

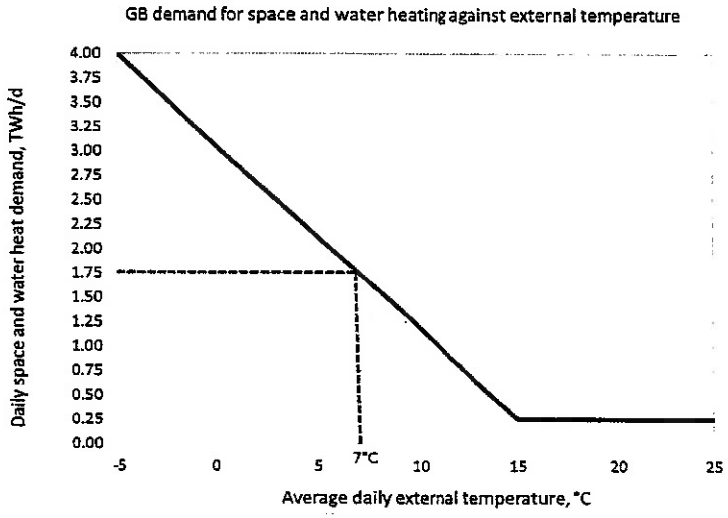


Figure 3.1: Daily GB demand for domestic space and water heating against average daily external temperature

Determine:

i) Average building space heat demand in MWh_{th} for the year

[1.5]

ii) Average water heat demand in MWh

[1]

iii) Average Heat Loss Parameter (HLP) in W/m^2K

[1.5]

c) Climate change is estimated to result in an increase in average temperatures of 2°C for the heating period by 2050. Assuming a 30% average improvement in building insulation, determine

i) Average building space heat demand in 2050 in MWh

[1.5]

ii) Total GB demand for domestic space and water heating

[1.5]

d) Using the above figure, plot the revised space and water heat demand for 2050 against average daily temperature stating any assumptions made.

[3]

e) Give a brief description of an air source heat pump (ASHP) and the four stages of the thermal cycle.

[3]

f) Over the next 30 years and on the assumption that 14 million buildings are converted from natural gas for heating to air source heat pumps

i) estimate the increase in electricity demand for space and water heating in 2050 based on your answer in part iii) b and stating any assumptions made.

[1]

ii) Determine from your figure the total daily heat demand when the average daily external temperature is -5°C and explain what happens to the efficiency of the air source heat pump at these lower temperatures.

[2]

iii) Calculate the daily electricity demand for an average building when the temperature is -5°C stating any assumptions made.

[1]

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4. Question on Wind Balancing, Hydro Power and Offshore Transmission

Wind Balancing

a) Increased penetration of variable and not fully predictable wind generation will increase requirements related to balancing of demand and supply. What will drive the amount of various forms of reserve services that will be needed to manage demand supply balance?

[3]

b) Consider a system with 23 GW of installed wind generation capacity in which demand varies between a minimum of 21 GW and a peak of 60 GW. The generation mix is composed of 7GW of inflexible nuclear plant capacity and conventional Combined Cycle Gas Turbine (CCGT) plant with characteristics presented in Table 4.1. CCGT plant emits 0.39 tonnes of CO_2 per MWh produced when operated at full output.

Technology	Rating of the unit [MW]	Minimum stable generation (MSG) [MW]	Marginal cost at full output [£/MWh]	Loss in efficiency when run at MSG [%]
CCGT	550MW	320MW	63	22

Table 4.1: Characteristics of CCGT plant

We will consider hourly cost of balancing demand and supply in this system during minimum and maximum demand conditions, assuming the output of wind generation of 15 GW. In order to cover the uncertainty in wind power output, the system operator decides to schedule 3910MW of reserve to be provided by part loading synchronised CCGTs. Assume that 7GW of inflexible nuclear plant will operate at all times.

i) Determine the minimum number of flexible generators that will need to run to provide the reserve required and their power output

[1]

ii) Determine the total generation costs incurred and the total CO_2 emitted by the CCGTs that provide required reserve

[2]

iii) Calculate the amount of wind generation that may need to be curtailed under both minimum and maximum demand conditions.

[2]

iv) If demand side response can provide 1380MW of reserve, determine the amount of wind power that may need to be curtailed at the minimum demand conditions.

[2]

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Hydro Power

c) Each unit of a reservoir based hydro power plant has a rated power capacity of 800 MW. The heights of different parts of the plant from a reference level are given in the Table 4.2:

Surface of headwater in reservoir	h_1	110 m
Inlet point into penstock	h_2	30 m
Turbine inlet and outlet	$h_3 = h_4$	0 m
Surface of tailwater in river below	h_5	10 m

Table 4.2: Heights of different parts of the hydro power plant from a reference level

The penstock has a constant inner diameter of 12 m. The combined efficiency of the turbine generator unit is 90%. The density of water is 1000 kg/m^3 and acceleration due to gravity is 9.81 m/s^2. Atmospheric pressure is 101.32 kPa. Neglect any loss of head.

i) How much volume flow rate or discharge (in m^3/s) of water is required for rated power output from the turbine generator unit?

[2]

ii) What is the velocity (m/s) and absolute pressure (kPa) of water at the inlet of the turbine?

[3]

Offshore Transmission

d) Justify the following statements giving appropriate reasons:

i) An LCC HVDC substation requires larger space than an equivalent VSC HVDC substation.

[2]

ii) Stronger and lighter polymeric cables can be used for VSC HVDC links but not for LCC HVDC links.

[2]

iii) For sub-sea cable transmission, the breakeven distance beyond which HVDC is more cost-effective than AC is larger in case of a remote offshore wind farm connection than a sub-sea interconnector between two onshore AC systems.

[1]